Topological - Aware Analysis of Tractography in Parkinson's Disease

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Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder marked by loss of dopaminergic neurons in the substantia nigra, with pathology that spreads through and disrupts structural brain networks. Detecting these alterations remains challenging with conventional graph-based analyses of diffusion MRI metrics, which primarily capture pairwise connections and neglect higher-order structural organization. To address this, we characterized the global and local topology of structural connectivity using **Topological Data Analysis (TDA).** By quantifying complex, multiscale network features, TDA provides richer structural representations that extend beyond standard dyadic interactions.

Our preliminary results showed a trend toward **depletion of high-order cliques** and **localized topological degradation** among core nodes of functional subnetworks in PD. We suggest that these topological features can provide informative structural priors for Graph-based machine learning methods like Graph Neural Networks (GNNs), enabling the discovery of complex connectivity patterns and ultimately support **more accurate and earlier diagnosis** of PD.



Methods

➢ Dataset

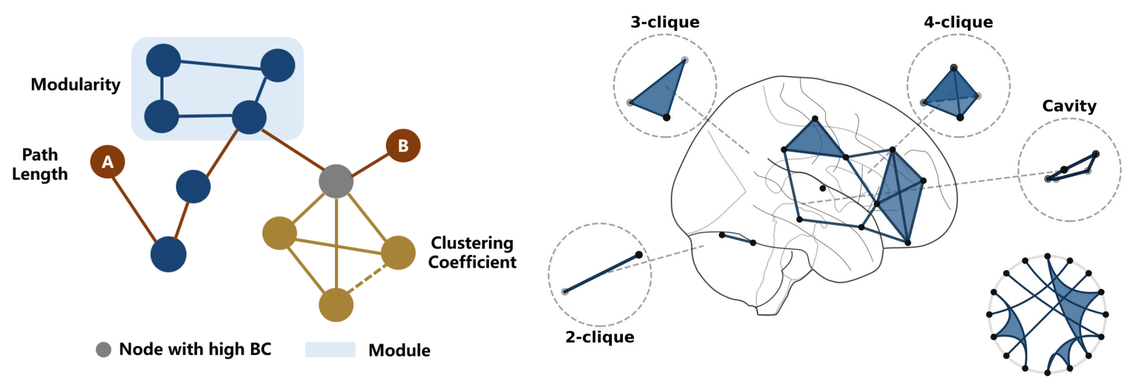
Data from the Parkinson' s Progression Markers Initiative (PPMI) includes PD patients (n = 15; 10M/5F; Age 60.2 ± 8.5) and age-matched Healthy Controls (HC, n = 15; 10M/5F; Age 60.2 ± 8.4).

➢ Network Construction: Structural Connectomes Pipeline

Structural connectomes are extracted from SIFT2-filtered and unfiltered tractograms, weighted by DTI metrics - streamline count (SC), fractional anisotropy (FA), mean diffusivity (MD), and radial/axial diffusivity (RD/AD).

➢ Graph-Theory Measures

- Local Level: Node strength, Betweenness Centrality, Clustering coefficient.
- Global Level: Global Efficiency, Clustering Coefficient, Modularity.



➢ TDA Measures

- Cliques: All-to-all connected subgraphs of any size.
- Cavities: Homology classes, intutitively: closed arrangements of interconnected cliques that encloses a void.
- Persistent Betti Curves: Track topological cavities (β_1 , β_2 , β_3).
- Node Participation: The total number of maximal cliques including a given node.

Main Results

I. Graph-Theory Measures are Insufficient to Uncover Brain Structure Alterations in PDs

➢ **Local Measures:** A few nodes (6-14) showed nominal significance (*Node Strength* here as an example).

➢ **Global Measures:** No statistically significant differences between PD and HC across all global metrics, indicating traditional pairwise metrics may lack the sensitivity to capture PD network alterations.

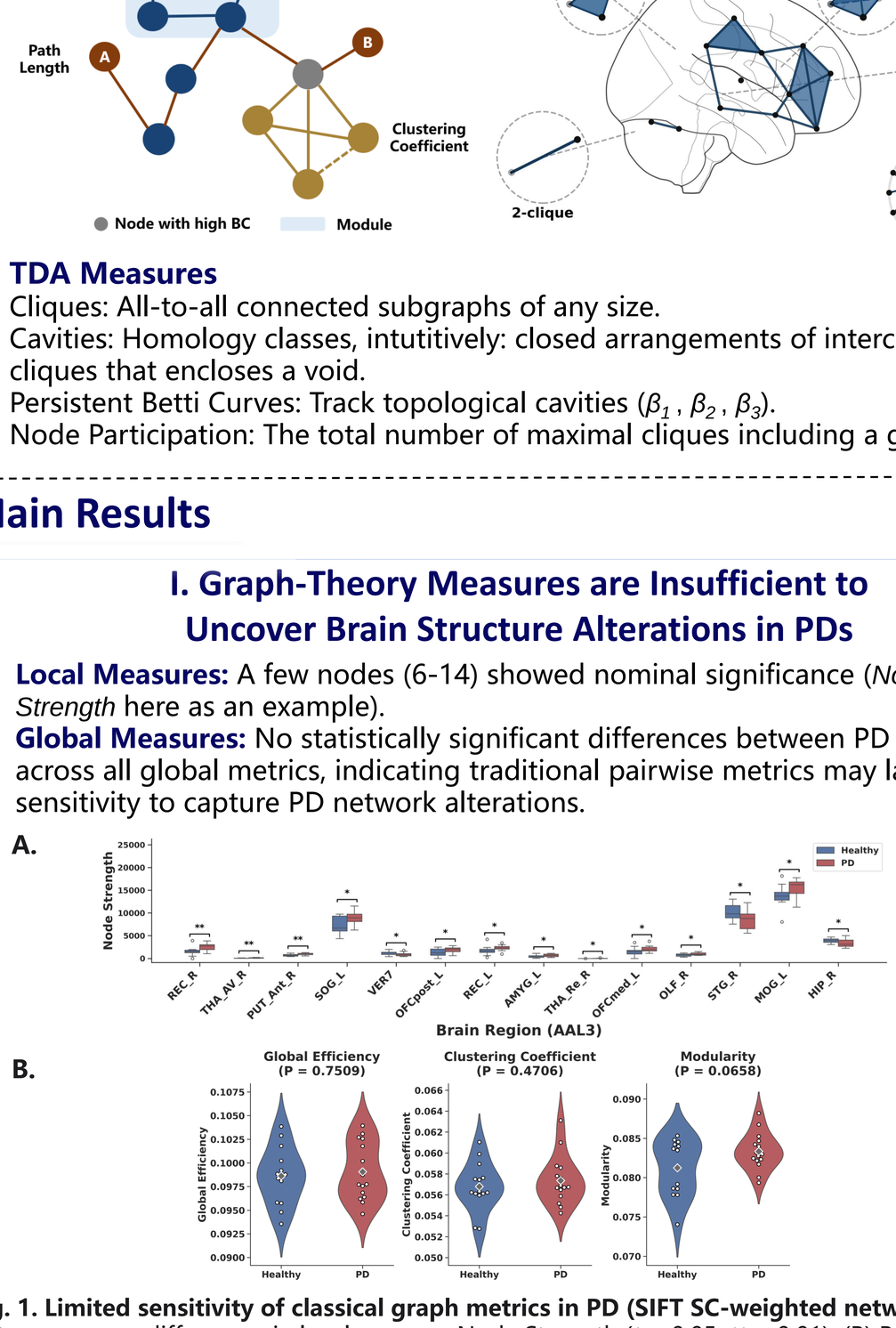


Fig. 1. Limited sensitivity of classical graph metrics in PD (SIFT SC-weighted networks). Between-group differences in local measure Node Strength (*p<0.05, **p<0.01). (B) Between-group differences in global measures.

II. TDA Captures High-Dimensional Topological Collapse and Network Fragmentation in PD

1. Clique Depletion Indicates Potential Topological Degradation

Structural fidelity was validated against Degree-preserving randomizations and Erdős-Rényi random networks. Compared with HCs, PD networks showed a trending depletion of high-dimensional cliques, indicating a potential degradation of global topological architectures.

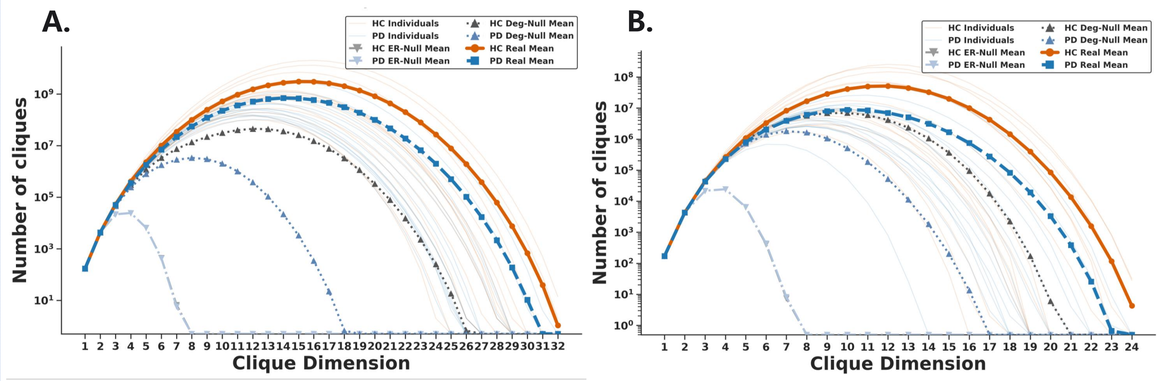


Fig. 2. Distribution of cliques in HCs, PDs and random networks, thresholded at an edge density of ρ = 0.3. (A. MD-weighted networks B. RD-weighted networks).

2. Increased Topological Cavities Reflect Network Fragmentation

Comparisons with random networks verified the authenticity of the network structure. Integrated Betti values in higher dimensions (β_2 and β_3) showed a significant increase along with the loss of dense cliques in PD patients, suggesting a potential fragmentation of the brain network into complex structural cavities and loops.

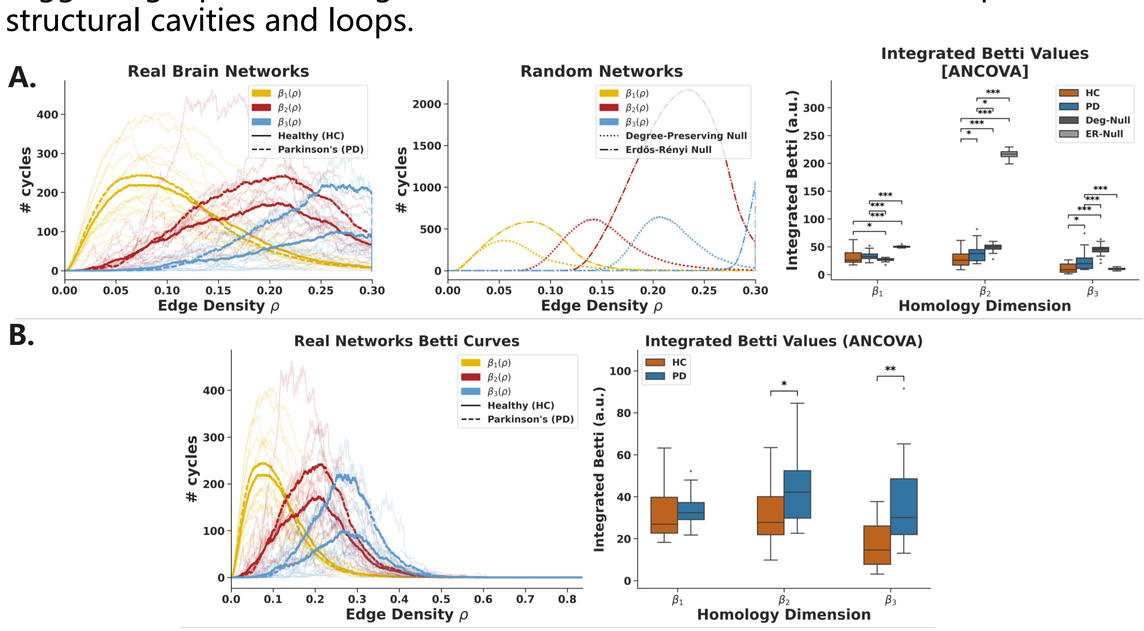


Fig. 3. Persistent Topological Cavities of HCs and PDs (RD-weighted networks). (A) Betti curves and integrated Betti values comparing real brain networks with random networks at an edge density of ρ = 0.3. (B) Full-density Betti curves and comparisons of integrated Betti values between HCs and PDs. (*p<0.05, **p<0.01, ***p<0.001)

3. PD Induces Localized Topological Collapse in Motor Subnetworks

➢ **Clique Shift:** PD networks shift from high- to low-dimensional cliques, indicating a collapse of local complex topologies.

➢ **Targeted Depletion:** Core motor subnetworks (e.g., *CBGT_Motor*, *SMN*, *Tremor_Cerebellar*) show significant loss of high-dimensional structures, aligning with clinical motor deficits.

➢ **Hub Vulnerability:** Strong correlations reveal that classical hubs anchor these complex cliques. Their collapse drives the overall topological degradation in PD.

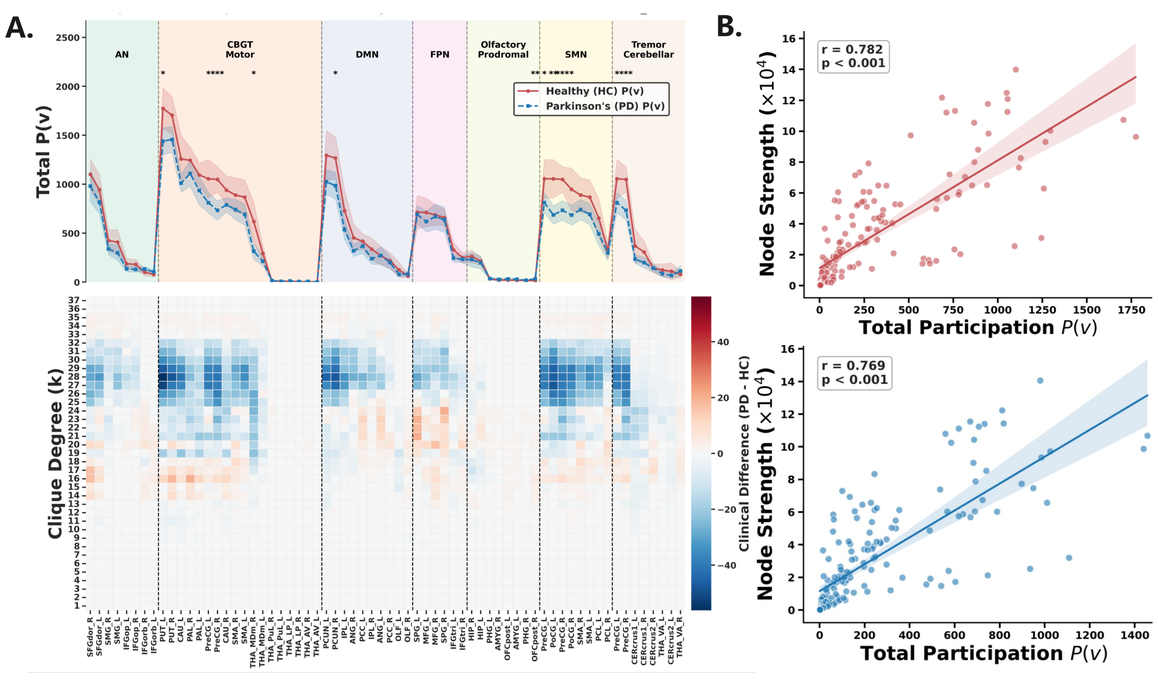


Fig. 4. Dimensional shift in node participation at an edge density of ρ = 0.3.**(SIFT SC-weighted networks)** (A) Node participation sorted by core subnetworks (*p<0.05, **p<0.01). (B) Correlations between Node Participation P(v) and Node Strength of HCs and PDs.

Summary

- **Limits of Classical Metrics:** Traditional pairwise metrics lack sensitivity to detect early, localized structural degradation in PD.
- **High-Order Fragmentation:** TDA reveals high-dimensional topological collapse, marked by depleted cliques and increased complex cavities (β_2 , β_3).
- **Localized Vulnerability:** Degradation specifically targets core motor circuits (e.g., *CBGT_Motor*, *SMN*, *Tremor_Cerebellar*) driven by structural shifts in network hubs.
- **Multi-Metric Synergy:** Complementary dMRI metrics collectively provide a comprehensive profile of pathological alterations.